

Omar Nabail

omargamal039@gmail.com

| github.com/OmarNabail

| linkedin.com/in/omar-nabail

+49 15225414448 | Germany

PROFILE

M.Sc. Artificial Intelligence & Robotics student and research assistant who builds, evaluates, and ships generative-AI systems end to end. Fine-tuned and deployed an LLM for text-to-visualization (97.0% code-execution rate), cut LLM input tokens by 90.2% in an industry collaboration, and benchmarked RAG pipelines on accuracy, latency, and cost. Strong in Python, LLMs, retrieval, multimodal evaluation, and production-minded delivery (FastAPI, Docker, CI/CD).

EXPERIENCE

NLLG Lab, UTN – Student Research Assistant

Jul 2026 – Present

- Contribute to research on generative multimodal models, covering evaluation methodology, failure-mode analysis and result reporting across NLP and vision-language projects.

UTN / Revel8 – Industry Collaboration: AI for Cybersecurity

Mar 2025 – Feb 2026

- Reduced LLM input tokens by 90.2% through MIME-aware preprocessing, improving efficiency, scalability, and expected inference cost for phishing analysis.
- Worked with industry stakeholders and a research team to turn a business problem into requirements, then built and evaluated a proof of concept combining local/cloud LLMs, prompt engineering (zero-shot, few-shot and chain-of-thought), and fine-tuned RoBERTa/EmoBERTa classifiers.

SELECTED AI & MACHINE LEARNING PROJECTS

PlotCraft: Fine-Tuned LLM for Text-to-Visualization

Python, PyTorch, Hugging Face, FastAPI

- Built an end-to-end system that turns plain-language requests into executable Python/Matplotlib visualizations, automating a repetitive chart-creation task; created PYVEC, a filtered dataset of 53,846 text-to-visualization examples.
- Fine-tuned Qwen2.5-3B with LoRA-based SFT followed by GRPO, reaching a 97.0% internal code-execution rate; analyzed failure cases to guide model and workflow improvements.
- Deployed a public ZeroGPU demo with a Gradio interface, containerized FastAPI service (health checks, configuration management, error handling), sandboxed Matplotlib preview with execution limits, automated tests, and CI/CD ([Live Demo](#); [GitHub](#)).

Agentic Scientific Figure Generation

Python, TikZ, LLMs, VLM Evaluation

- Designed and implemented a multi-stage workflow that interprets a natural-language request, routes it to Python/Matplotlib, TikZ, or direct-image generation, selects among candidates, and validates results with execution checks and visual evaluation.
- Investigated cross-representation training and routing; evaluated outputs with execution success, visual similarity, VLM judges, and human evaluation to determine where automation can improve R&D content-creation productivity without sacrificing quality.

Local RAG with Hybrid Retrieval & Reranking *Ollama, LlamaIndex, ChromaDB, Sentence Transformers*

- Built a local RAG application over product documentation and historical support tickets, covering document processing, indexing, retrieval, answer generation, and evaluation with Ragas.
- Benchmarked BM25, dense-embedding, hybrid, and cross-encoder reranking retrieval on a hand-labelled query set using Recall@k and MRR; quantified trade-offs between retrieval accuracy, latency, and computational cost.
- Developed a Streamlit dashboard for comparing retrieval configurations side by side.

Multilingual LLM Evaluation for Content Moderation *LLM APIs, Python, Data Analysis*

- Compared GPT-4, Claude, and Qwen on hate-speech moderation across languages, investigating consistency, robustness, and fairness differences through a structured, reproducible evaluation suitable for model-selection decisions.
- Built data-processing and evaluation pipelines and produced transparent evaluation reports communicating failure patterns.

ADDITIONAL ML & COMPUTER VISION PROJECTS

Outlier Exposure for Valve Condition Monitoring *Python, PyTorch, MobileFaceNet, AdaCos*

- Developed an acoustic anomaly-detection system for valve condition monitoring, converting audio into frequency-domain (DFT) and spectrogram representations and learning embeddings with MobileFaceNet and sub-cluster AdaCos.
- Investigated outlier exposure under domain shift; achieved 81.38% AUC and 72.29% partial AUC.

CAPTCHA Recognition with a CNN-Transformer Hybrid *Python, Deep Learning, CNNs, Transformers*

- Designed a hybrid architecture for CAPTCHA text recognition combining deformable convolutions with triplet attention; placed third among eight teams in the UTN 2025 Computer Vision Challenge.

German Trash Classification with CLIP *Python, OpenAI CLIP, Computer Vision*

- Developed a zero-shot waste-classification application with German and multilingual user interaction, integrated through an API; achieved 95.4% Top-5 accuracy on real-world waste images without training a dedicated classifier per category.

SOFTWARE & SYSTEMS PROJECTS

Performance Analysis System for Cluster Hosts and Clients *Kafka, DuckDB, LZ4*

- Developed decentralized monitoring pipelines for distributed cluster analytics, streaming performance data with Kafka and processing it with DuckDB.
- Implemented batching and LZ4 compression, and separated customer and operator information into distinct encrypted data streams.

Airline Reservation System *JavaScript, Node.js, React, MongoDB*

- Developed a full-stack flight-search and reservation application with a REST API (Node.js, Express), MongoDB backend, and React/Material UI frontend.
- Implemented registration, login with encrypted password storage, flight booking, and automated email notifications via Nodemailer.

Coast Guard AI Rescue Simulation *Java, Search Algorithms, Java Swing*

- Built an autonomous agent navigating a dynamic sea grid with sinking ships, passenger rescue, black-box recovery, and time-dependent consequences.
- Implemented eight search strategies (including BFS, DFS, iterative deepening, uniform-cost, greedy, and A*) with custom heuristics, and added a Swing interface for visualizing missions.

Dance Floor: Interactive 3D Application

C++, OpenGL, 3D Graphics

- Built an interactive real-time 3D environment with a controllable character (WASD movement) and multiple dance animations, side, front, and free camera modes, custom 3D mesh loading, and textured-model rendering.

ShootAliens: 2D Shooter Game

C++, OpenGL, GLUT

- Developed a complete two-level 2D shooter from scratch with mouse-directed shooting, randomized enemy spawning, collision detection, a power-up mechanic, and GLUT-timer-driven animation, progression, and win/lose states.

Autonomous Car

C/C++, Arduino, Processing, Fritzing

- Designed and built a physical autonomous vehicle; programmed motor-control and sensor-integration logic in C/C++ for an Arduino microcontroller and designed the circuit in Fritzing.
- Built a Processing-based serial interface for real-time monitoring and documented the system architecture, components, and behaviour.

EDUCATION

UTN – M.Sc. Artificial Intelligence & Robotics

Sep 2024 – Present

- **Relevant Coursework:** Multimodal Foundation Models, Natural Language Generation and Evaluation, Deep Learning, Computer Vision, Machine Learning, Artificial Intelligence, Cloud Databases, Data Engineering, Working Effectively with AI Tools

German University in Cairo – B.Sc. Computer Science and Engineering

Sep 2017 – Sep 2023

- **Bachelor's Thesis: Outlier Exposure for Valve Condition Monitoring**

SKILLS

- **Generative AI & LLMs:** LLM APIs, prompt engineering (zero-shot, few-shot, chain-of-thought), fine-tuning (LoRA, SFT, GRPO), RAG, retrieval, reranking, agentic and multi-stage workflows
- **ML & Deep Learning:** PyTorch, Hugging Face Transformers, Sentence Transformers, RoBERTa, CLIP, CNNs, Transformers, BM25, embeddings, LlamaIndex, ChromaDB, Ollama, Ragas
- **Evaluation & Communication:** benchmark design, failure-mode analysis, automated, VLM-based, and human evaluation, stakeholder communication
- **MLOps & Delivery:** FastAPI, REST APIs, Docker, GitHub Actions, CI/CD, pytest, Git, Linux, Hugging Face Spaces, Gradio, Streamlit
- **Programming & Data:** Python, SQL, Java, C/C++, JavaScript, NumPy, Matplotlib, TikZ, DuckDB, Kafka, JSON
- **Web & Embedded:** Node.js, Express, React, MongoDB, OpenGL, Arduino, Processing, Fritzing

LANGUAGES

English: Fluent (C1) **German:** Intermediate (B1) **Arabic:** Native (C2)